Lecture 10: Conditional Probability

First example: student and parent drug use

		parents		
		used	not	Total
student	uses	125	94	219
	not	85	141	226
	Total	210	235	445

Random phenomenon: sample one student from this group and ask about drug use.

Venn diagram representation:

- If the student tells us their parents use(d) drugs, what is the probability that they also use?

- If the student tells us they do not use drugs, what is the probability that their parents did/do?

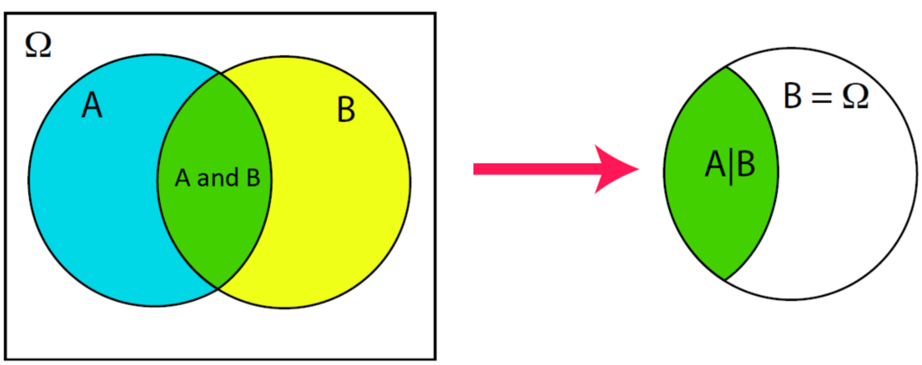
Terminology: Marginal and Joint Probabilities

Marginal probability:

Joint probability:

Calculating marginal probabilities from joint probabilities: **Law of Total Probability**

Conditional Probability



If events A and B are independent, then ...

Example: Smallpox Outbreak in Boston, 1721

		inoculated		
		yes	no	Total
result	lived	238	5136	5374
	died	6	844	850
	Total	244	5980	6224

Random phenomenon: sample one Bostonian's record and check how they fared during the outbreak

- What is the probability that they were not inoculated and died of smallpox?

- If they were not inoculated, what is the probability that they died of smallpox?

- If they were inoculated, what is the probability that they died of smallpox?

Random phenomenon: Take a SRS of size 2 from Alice, Bob, Cindy, Dave, Eva

Recall sample space:

- $P(B) =$

- $P(B|A) =$

General Multiplication Rule

Example: Suppose we are given only two pieces of information from the smallpox data: 96.08% of residents were not inoculated, and 85.88% of the residents who were not inoculated ended up surviving. How could we compute the probability that a resident was not inoculated and survived?

Rules for conditional probabilities:

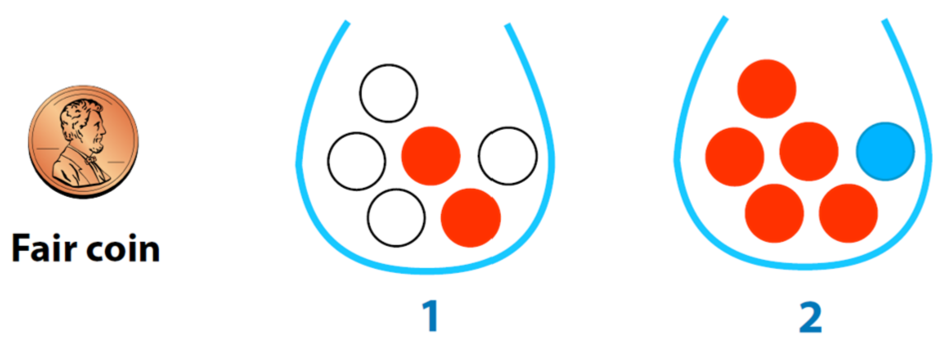
- If A and B are disjoint events and C is another event,

- For any two events A and B,

Working off of the previous question, what is the probability that a randomly chosen resident died given they were not inoculated?

Bayes' Theorem

Random phenomenon: Flip a fair coin to determine which jar (heads = jar 1, tails = jar 2) we will draw a colored ball from without looking and record the color of the ball that is drawn.



Sample space?

Working out probabilities for each outcome: probability tree

Given that we drew a red ball, what is the probability that the coin flip came up heads?

Bayes' Theorem: general form

Example: Monty Hall Problem

You're on a game show where you're given the choice of three doors: Behind one door is a car; behind the others, goats (obviously you want the car). You start the game by choosing a door, say door number 1. To make things interesting, the host, who knows what's behind each door, opens one of the doors you didn't choose, say door number 3, revealing a goat. He then asks, "Do you want to switch your choice to door number 2, or stick with door number 1?" What should you do?

Practice: Spam-B-Gone is a computer application that learns to sort e-mail into 3 distinct categories — high-priority (**H**), low-priority (**L**), or spam (**S**) — based on feedback collected from a user over a week-long trial period. During Bill's trial period, he received a total of 1200 e-mails and labeled 200 of them as **H**, 600 as **L**, and 400 as **S**. Spam-B-Gone now hopes to use Bill's feedback to automatically classify all future e-mails he receives.

One of Spam-B-Gone's classification methods involves scanning for certain keywords, e.g. "offer", that are more likely to show up in spam e-mails. In Bill's case, just 4 of the 200 e-mails he labeled **H** included the word "offer", while 30 of the 600 **L** e-mails and 320 of the 400 **S** emails included "offer".

- If all Bill's future e-mails follow the same distribution as we observed during the trial period, what is the probability that the next e-mail Bill receives is not spam?

- What is the probability that the next e-mail Bill receives includes the word "offer"?

- Given that the next e-mail Bill receives includes the word "offer", what is the probability that he would classify it as a spam e-mail?